

for some positive constant $\tilde{c}_0$, where

$$\sigma_{\lambda,1,1}^2 := \text{var}_{X,Y}(\mathbb{E}_{X',Y}[h_\lambda(X, X', Y, Y')])$$

and

$$\sigma_{\lambda,2,2}^2 := \text{var}_{X,X',Y,Y'}(h_\lambda(X, X', Y, Y')) \leq \frac{16M\kappa_2}{\lambda_1 \cdots \lambda_d}$$

as shown earlier. Using the above results, we get

$$\begin{aligned} \sigma_{\lambda,1,1}^2 &:= \text{var}_{X,Y}(\mathbb{E}_{X',Y}[h_\lambda(X, X', Y, Y')]) \\ &= \text{var}_{X,Y}(G_\lambda(X) - G_\lambda(Y)) \\ &\leq \mathbb{E}_{X,Y}[(G_\lambda(X) - G_\lambda(Y))^2] \\ &\leq 2(\mathbb{E}_X[G_\lambda(X)^2] + \mathbb{E}_Y[G_\lambda(Y)^2]) \\ &\leq 4M\|\psi * \varphi_\lambda\|_2^2. \end{aligned}$$

Letting $\tilde{C}_1(M, d) := 4 \max\{4\tilde{c}_0 M, 16\tilde{c}_0 M \kappa_2\}$, we deduce that

$$\begin{aligned} \text{var}_{p \times q}(\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n)) &\leq \tilde{c}_0 \left(\frac{\sigma_{\lambda,1,1}^2}{n} + \frac{\sigma_{\lambda,2,2}^2}{n^2} \right) \\ &\leq \frac{1}{4} \tilde{C}_1(M, d) \left(\frac{\|\psi * \varphi_\lambda\|_2^2}{n} + \frac{1}{n^2 \lambda_1 \cdots \lambda_d} \right) \\ &\leq \tilde{C}_1(M, d) \left(\frac{\|\psi * \varphi_\lambda\|_2^2}{2n} + \frac{1}{(2n)^2 \lambda_1 \cdots \lambda_d} \right). \end{aligned}$$

E.4 Proof of Proposition 4

Recall that $(\widehat{M}_\lambda^b)_{1 \leq b \leq B}$ is defined in Sections 3.2.1 and 3.2.2 for the estimators $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2$ and $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$, respectively, and that $\widehat{M}_\lambda^{B+1} := \widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n)$ for both estimators. Let us recall that the $(1-\alpha)$ -quantile function of a random variable X with cumulative distribution function F_X is given by

$$q_{1-\alpha} = \inf\{x \in \mathbb{R} : 1 - \alpha \leq F_X(x)\}.$$

We denote by F_B and F_{B+1} the empirical cumulative distribution functions of $(\widehat{M}_\lambda^b)_{1 \leq b \leq B}$ and $(\widehat{M}_\lambda^b)_{1 \leq b \leq B+1}$, respectively.

For the case of the estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2$, we denote by F_∞ the cumulative distribution function of the conditional distribution of $\widehat{M}_\lambda^\sigma$ (defined in Equation (10)) given $\mathbb{X}_m$ and $\mathbb{Y}_n$, where the randomness comes from the uniform choice of permutation σ among all possible permutations of $\{1, \dots, m+n\}$. For the case of the estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$, we similarly denote by F_∞ the cumulative distribution function of the conditional distribution of $\widehat{M}_\lambda^\epsilon$ (defined in Equation (11)) given $\mathbb{X}_m$ and $\mathbb{Y}_n$, where the randomness comes from the n i.i.d. Rademacher variables $\epsilon := (\epsilon_1, \dots, \epsilon_n)$ with values in $\{-1, 1\}^n$.

Based on the above definitions, we can write

$$\hat{q}_{1-\alpha}^{\lambda,\infty}(\mathbb{X}_m, \mathbb{Y}_n) = \inf\{u \in \mathbb{R} : 1 - \alpha \leq F_\infty(u)\}$$

and

$$\begin{aligned} \hat{q}_{1-\alpha}^{\lambda,B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) &= \inf\{u \in \mathbb{R} : 1 - \alpha \leq F_{B+1}(u)\} \\ &= \inf\left\{u \in \mathbb{R} : 1 - \alpha \leq \frac{1}{B+1} \sum_{b=1}^{B+1} \mathbb{1}(\widehat{M}_\lambda^b \leq u)\right\} \\ &= \inf\left\{u \in \mathbb{R} : (B+1)(1 - \alpha) \leq \sum_{b=1}^{B+1} \mathbb{1}(\widehat{M}_\lambda^b \leq u)\right\} \\ &= \widehat{M}_\lambda^{\bullet[(B+1)(1-\alpha)]} \end{aligned}$$

where $\widehat{M}_\lambda^{\bullet 1} \leq \dots \leq \widehat{M}_\lambda^{\bullet B+1}$ denote the ordered simulated test statistics $(\widehat{M}_\lambda^b)_{1 \leq b \leq B+1}$.

Now, for any given $\delta > 0$, define the event

$$\mathcal{A} := \left\{ \sup_{u \in \mathbb{R}} |F_B(u) - F_\infty(u)| \leq \sqrt{\frac{1}{2B} \log\left(\frac{4}{\delta}\right)} \right\}.$$

As noted by Kim et al. (2022, Remark 2.1), Dvoretzky–Kiefer–Wolfowitz inequality (Dvoretzky et al., 1956), more precisely the version with the tight constant which is due to Massart (1990), then guarantees that $\mathbb{P}_r(\mathcal{A} | \mathbb{X}_m, \mathbb{Y}_n) \geq 1 - \frac{\delta}{2}$ for any $\mathbb{X}_m$ and $\mathbb{Y}_n$, so we deduce that $\mathbb{P}_{p \times q \times r}(\mathcal{A}) \geq 1 - \frac{\delta}{2}$. We now assume that the event $\mathcal{A}$ holds, so the bound we derive holds with probability $1 - \frac{\delta}{2}$. Notice that we cannot directly apply the Dvoretzky–Kiefer–Wolfowitz inequality to F_{B+1} since it is not based on i.i.d. samples. Nevertheless, under the event $\mathcal{A}$, we have

$$\begin{aligned} \hat{q}_{1-\alpha}^{\lambda,B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) &= \inf\{u \in \mathbb{R} : 1 - \alpha \leq F_{B+1}(u)\} \\ &= \inf\left\{u \in \mathbb{R} : 1 - \alpha \leq \frac{1}{B+1} \sum_{b=1}^{B+1} \mathbb{1}(\widehat{M}_\lambda^b \leq u)\right\} \\ &\leq \inf\left\{u \in \mathbb{R} : 1 - \alpha \leq \frac{1}{B+1} \sum_{b=1}^B \mathbb{1}(\widehat{M}_\lambda^b \leq u)\right\} \\ &= \inf\left\{u \in \mathbb{R} : (1 - \alpha) \frac{B+1}{B} \leq F_B(u)\right\} \\ &\leq \inf\left\{u \in \mathbb{R} : (1 - \alpha) \frac{B+1}{B} \leq F_\infty(u) - \sqrt{\frac{1}{2B} \log\left(\frac{4}{\delta}\right)}\right\} \\ &= \inf\left\{u \in \mathbb{R} : \underbrace{(1 - \alpha) \frac{B+1}{B} + \sqrt{\frac{1}{2B} \log\left(\frac{4}{\delta}\right)}}_{:= 1 - \alpha^*} \leq F_\infty(u)\right\} \\ &= \hat{q}_{1-\alpha^*}^{\lambda,\infty}(\mathbb{X}_m, \mathbb{Y}_n). \end{aligned}$$

Now, we take B large enough (only depending on α and δ) such that

$$(1 - \alpha) \frac{B + 1}{B} + \sqrt{\frac{1}{2B} \log \left(\frac{4}{\delta} \right)} \leq 1 - \frac{\alpha}{2}$$

so that $\hat{q}_{1-\alpha}^{\lambda, \infty}(\mathbb{X}_m, \mathbb{Y}_n) \leq \hat{q}_{1-\alpha/2}^{\lambda, \infty}(\mathbb{X}_m, \mathbb{Y}_n)$ under the event $\mathcal{A}$. By reducing this problem to a quadratic equation with respect to B , we find

$$B \geq \frac{2}{\alpha^2} \left(\frac{1}{2} \ln \left(\frac{4}{\delta} \right) + \alpha - \alpha^2 + \sqrt{\left(\frac{1}{2} \ln \left(\frac{4}{\delta} \right) + \alpha - \alpha^2 \right)^2 - \alpha^2(1 - \alpha)^2} \right).$$

In particular, by upper bounding $-\alpha^2(1 - \alpha)^2$ by 0, we find that the above inequality holds as soon as

$$B \geq \frac{4}{\alpha^2} \left(\frac{1}{2} \ln \left(\frac{4}{\delta} \right) + \alpha(1 - \alpha) \right).$$

Note that this condition is in particular trivially satisfied if

$$B \geq \frac{3}{\alpha^2} \left(\ln \left(\frac{4}{\delta} \right) + \alpha(1 - \alpha) \right).$$

With this choice of B , we have

$$\mathbb{P}_{p \times q \times r} \left(\hat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) \leq \hat{q}_{1-\alpha/2}^{\lambda, \infty}(\mathbb{X}_m, \mathbb{Y}_n) \right) \geq 1 - \frac{\delta}{2}.$$

We now upper bound $\hat{q}_{1-\alpha}^{\lambda, \infty}(\mathbb{X}_m, \mathbb{Y}_n)$ for the two estimators $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2$ and $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$ separately. We then use this to prove the required upper bound on $\hat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n)$.

Quantile bound for MMD estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2$ defined in Equation (3).

In this case, we base our reasoning on the work of Kim et al. (2022, proof of Lemma C.1). Recall from Equation (7) that we assume that $m \leq n$ and $n \leq Cm$ for some positive constant C . We use the notation presented in Section 3.2.1 that $U_i := X_i$ and $U_{m+j} := Y_j$ for $i = 1, \dots, m$ and $j = 1, \dots, n$. By the result of Kim et al. (2022, Equation 59), there exists some $c_1 > 0$ such that

$$\hat{q}_{1-\alpha}^{\lambda, \infty}(\mathbb{X}_m, \mathbb{Y}_n) \leq c_1 \sqrt{\frac{1}{m^2(m-1)^2} \sum_{1 \leq i \neq j \leq m+n} k_\lambda(U_i, U_j)^2 \ln \left(\frac{1}{\alpha} \right)}$$

almost surely. As shown in Equation (22), for the constant $\kappa_2(d)$ defined in Equation (21), we have

$$\max \{ \mathbb{E}_{X, X'} [k_\lambda(X, X')^2], \mathbb{E}_{X, Y} [k_\lambda(X, Y)^2], \mathbb{E}_{Y, Y'} [k_\lambda(Y, Y')^2] \} \leq \frac{M \kappa_2}{\lambda_1 \cdots \lambda_d} \quad (23)$$